COLUMBIA UNIVERSITY
School of Professional Studies

Neighborhood Characteristics and Their Relationship with Restaurant Health Inspection Outcomes in NYC

August 2026

Prepared by Aziza, Tristan, Nicole, Yuchun

APANPS5205_D01_2026_2 - APPLIED MACHINE LEARNING II

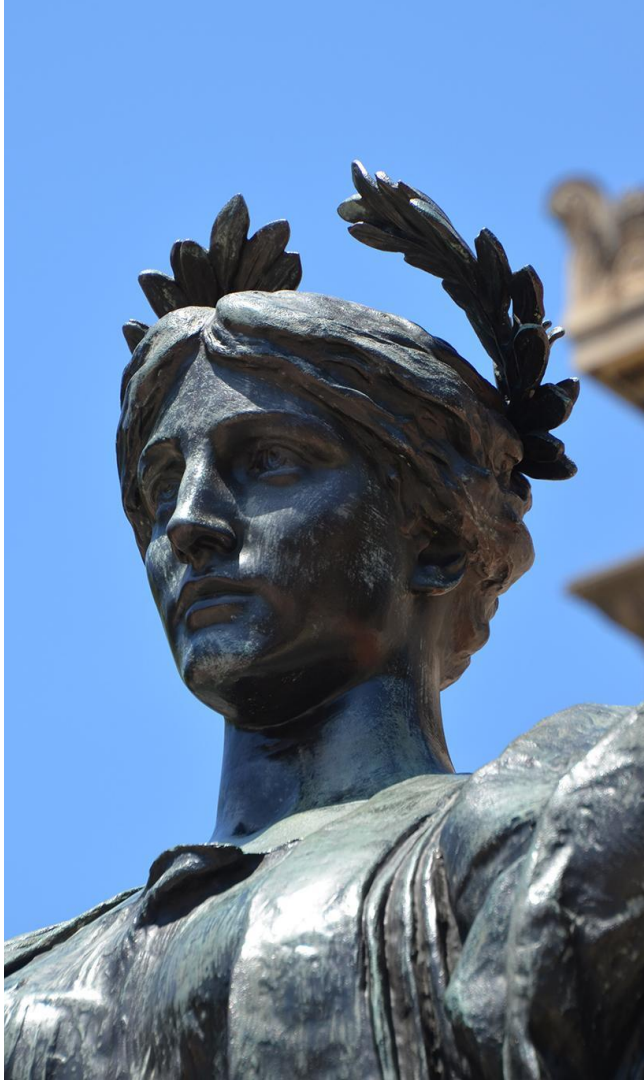
Table of Contents

3. Background and Problem Statement
4. Research Questions
5. Data Sources
6. Variables
7. Data Preparation
8. Preliminary Exploration
9. Planned Analysis
10. Expected Contributions
11. Limitations
12. Conclusion

Background and Problem Statement

New York City restaurants receive health inspection scores and letter grades based on food safety violations. While these results mainly reflect conditions inside each restaurant, neighborhood sanitation, population density, and socioeconomic factors may also be related to inspection performance. This project combines restaurant, 311, and demographic data to examine these relationships.

Our Solution: Combine restaurant inspection records, sanitation-related 311 complaints, and neighborhood demographic data to examine whether broader neighborhood conditions are associated with restaurant inspection scores and letter grades.



COLUMBIA SPS

Research Questions - Aziza

01

Do restaurant inspection scores and letter grades differ across the five New York City boroughs?

02

Is there a relationship between neighborhood sanitation conditions, measured through sanitation-related 311 service requests, and restaurant inspection outcomes?

03

Are neighborhood demographic characteristics, such as median household income, poverty rate, population density, and educational attainment, associated with restaurant inspection scores?

04

Which neighborhood characteristics appear to be the strongest predictors of restaurant inspection performance?

Data Source 1: Restaurant Inspections - Aziza

296,044
Raw Violation Records

59,841
Eligible Inspections

21,472
Final Restaurants

Dataset Source

DOHMH New York City Restaurant Inspection Results

Unit of Analysis

One row per restaurant using its most recent Cycle Inspection from January 2023 through June 2026.

: Lower scores indicate fewer violation points and better inspection performance.

Main Variables

- **Restaurant & CAMIS ID**
- **Borough & ZIP Code**
- **Cuisine Type**
- **Inspection Date & Type**
- **Inspection Score**
- **Grade & Violations**

Data Sources 2 & 3: Sanitation & Demographics - Aziza

Data Source 2

NYC 311 Requests

- **21.7M** raw service requests
- **1.15M** sanitation complaints
  - Rodents (tracked separately)
  - Dirty & unsanitary conditions
  - Dumping & missed collections

Aggregated by ZIP code

Data Source 3

ACS & Census Data

Demographic Predictors:

- Population & Density
- Median Household Income
- Poverty Rate
- Educational Attainment

Integration

Data Merging

Merge Key:

ZIP Code / ZCTA

Enables spatial matching between localized sanitation complaints, census demographic profiles, and restaurant business locations.

Combined Dataset Creation

Restaurant Inspections + 311 Sanitation Data + ACS Demographics ➔ Combined Restaurant-Level Dataset

Variables and Data Model - Tristan

Restaurant-Level Dataset

21,472 rows x 24 columns

One row per restaurant primary analysis file

Identifiers and Attributes

- CAMIS ID & restaurant name
- Borough and ZIP code
- Cuisine type
- Inspection date and type

Outcome Variables

- score (violation points)
- grade and grade_from_score
- is_grade_A indication
- n_violations / n_critical_violations

Neighborhood ZIP-Level Dataset

217 rows x 14 columns

One row per ZIP/borough aggregation and robustness checks

Neighborhood Sanitation (ZIP)

- sanitation_compliants
- rodent_complaints
- sanitation_per_1000
- rodent_per_1000

Neighborhood Demographics (ZIP)

- population & pop_density_sqmi
- median_hh_income
- poverty_rate
- pct_bachelors_plus

Data Model: many restaurants -> one ZIP-level neighborhood profile
Merge key: 5-digit ZIP code (Census ZCTA). Every restaurant record carries its ZIP's sanitation and demographic values; zip_avg_score summarizes restaurant outcomes back at the neighborhood level for robustness checks.

Data Preparation and Integration - Tristan

1. Clean and Filter

Remove malformed and placeholder records; standardize ZIP codes and dates; restrict to comparable Cycle Inspections, January 2023-June 2026

2. Collapse and Aggregate

Violation level rows -> inspection level -> most recent inspection per restaurant; 311 requests keyword matched and summed by ZIP code

3. Derive Variables

grade_from_score from the official 13/27 cutoffs; is_grade_A flag; sanitation and rodent complaints per 1,000 residents; ZIP-level zip_avg_score

4. Merge on ZIP Code

Left join restaurant + 311 sanitation + ACS demographics on 5-digit ZIP/ZCTA; missing 311 counts filled with 0, demographic gaps kept as missing

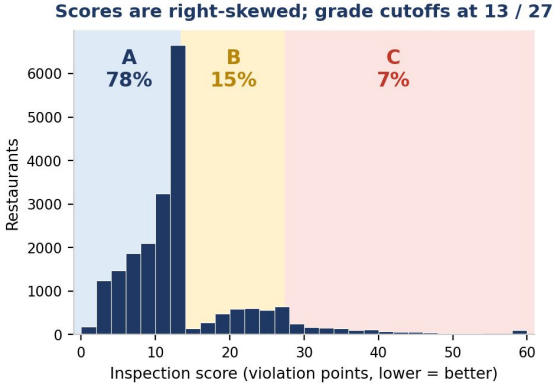
Final Cleaned Outputs, Ready for Analysis
21,472 rows x 24 columns (restaurant-level) 217 rows x 14 columns (ZIP-level) – reproducible pandas pipeline from raw NYC Open Data
Processed in memory-safe chunks: 296K violation records, 21.7M raw 311 requests, and 4 ACS tables merged without loading any full raw file into memory at once.

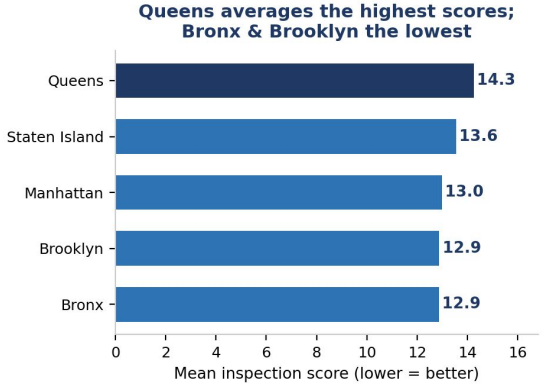
Preliminary Data Exploration - Yuchun

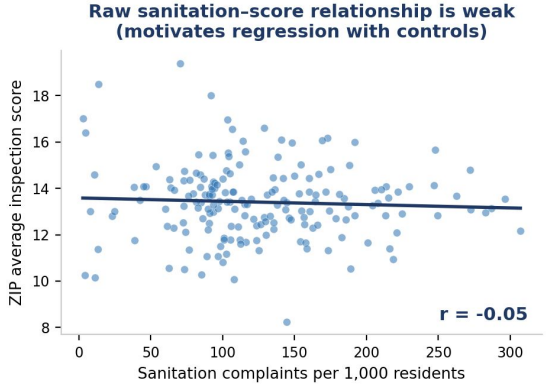
21,472 restaurants (one row each)

217 ZIP codes covered

24 analysis variables







- **Data quality:** 12.5% of posted letter grades are pending, so grades are derived from scores using the official 13 / 27 cutoffs (0% missing).
- **Coverage:** 254 restaurants (1.2%) sit in non-residential ZIPs (office towers, airports) and are flagged; 311 categories matched by keyword after the 2026 schema rename.

Planned Statistical Analysis - Yuchun

Unit of analysis: each restaurant's most recent cycle inspection (n = 21,472), with ZIP-level sanitation and demographic measures merged on.

01 Borough comparison

One-way ANOVA and Kruskal–Wallis tests of score and grade differences across the five boroughs.

→ **Answers RQ1**

02 Correlation analysis

Pearson and Spearman correlations between neighborhood measures and inspection outcomes; multicollinearity screen.

→ **Answers RQ2–RQ3**

03 Multiple linear regression

Score modeled on sanitation and demographics with borough and cuisine controls; ZIP-clustered standard errors.

→ **Answers RQ2–RQ3**

04 Logistic regression

Probability of an A grade (score-derived) on the same predictors — the outcome the public actually sees.

→ **Answers RQ2–RQ3**

05 Feature importance

Standardized coefficients compared with random-forest importance to rank neighborhood predictors.

→ **Answers RQ4**

06 Geographic visualization

Maps of residuals and predictions to reveal spatial patterns the predictors leave unexplained.

→ **Supports all RQs**

Robustness: analyses repeated at ZIP level (n = 217) · Results interpreted as associations, not causal effects

Expected Contributions and Limitations - Yuchun

EXPECTED CONTRIBUTIONS

Policy targeting

If sanitation conditions track poorer inspection outcomes, agencies can target sanitation services, rodent mitigation, and inspection resources to higher-risk areas.

Filling a research gap

Directly links restaurant inspection records with neighborhood environmental and socioeconomic data — a combination rarely studied together.

Reproducible open-data pipeline

A documented raw-to-clean workflow merging three public datasets by ZIP code, extensible to other cities and public-health questions.

LIMITATIONS

- ZCTAs approximate USPS ZIP codes; boundaries do not align perfectly.
- 311 complaint volume partly reflects residents' reporting propensity, not only true conditions.
- Observational, aggregated data: associations only — neighborhood-level results are not claims about individual restaurants (ecological fallacy).
- Non-residential ZIPs (1.2% of restaurants) excluded from neighborhood analyses; pending grades handled via score-derived grades.

Dataset ready for analysis: 21,472 restaurants · 217 ZIP codes · 98.8% with complete neighborhood measures · Jan 2023 – Jun 2026

Conclusion and Future Recommendation

CONCLUSION

- Restaurant inspection performance may vary across New York City neighborhoods.
- Combining restaurant inspection, sanitation, and demographic data provides a more complete understanding of food safety patterns.
- The analysis will identify which neighborhood characteristics are most strongly associated with inspection outcomes.

FUTURE RECOMMENDATIONS

- Include additional years of inspection and 311 data.
- Examine changes in inspection performance over time.
- Add more detailed geographic measures.
- Study specific restaurant or cuisine categories.
- Create an interactive dashboard or map showing neighborhood patterns.

COLUMBIA SPS

